



ARKA MANNA

**B.Tech in Mechanical Engineering
(2023–2027)**

CONTACT INFORMATION

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LINKEDIN PROFILE

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PORTFOLIO

<https://arka-manna.github.io/>

PROJECT SNAPSHOT (DETAILS ON PAGE 2)

- PRISMATIC – Mobile Inspection Robot
- Fixed-Wing UAV Platform (Boeing Competition)
- Hovercraft – Mechanical Design Prototype
- Single-Axis Helicopter Prototype

EDUCATIONAL HISTORY

- Completed I.S.C in Science from Oxford High School, Howrah in 2022
- B. Tech in Mechanical Engineering Heritage Institute of Technology-Kolkata, 2023-2027, 6th semester ongoing

PROFILE

Mechanical Engineering undergraduate with hands-on experience in **designing, fabricating and testing mechanical & mechatronic systems**. Worked on **mobile inspection robots, UAV platforms, and automation prototypes** involving **mechanical design, embedded control**, and **system integration**. Seeking exposure to heavy engineering, manufacturing, and R&D environments.

RELEVANT SKILLS

- **Design & Simulation:** SolidWorks, Ansys Fluent, Processing 4, MATLAB.
- **Robotics & Automation:** Robot Operating System (ROS), Gazebo.
- **Programming & Electronics:** Python, C++, Arduino, PCB Design.
- **IoT & Computer Vision:** OpenCV, ESP32 Wireless Communication.

ACHIEVEMENTS

- **1st Place** | CRYPTONIX-V1 – Cybersecurity Prototype Champion (Heritage Institute of Technology).
- **1st Place** | Sustainovation – Hardware Track Winner for Sustainable Tech (Amity/INTIN).
- **Top 30 National Finalist** | Kshitij (IIT Kharagpur) Boeing National Aeromodelling Competition.

WORK EXPERIENCE

- **Think Now Classroom | Product Developer**
Engineered 5+ STEM-aligned science kits, resulting in a 20% increase in student engagement during pilot testing
- **ASME HITK Student Section | Additional Vice-Chair**
Coordinated 3+ regional engineering workshops for 200+ students, securing 2 industries sponsorships

PROJECTS

PRISMATIC – MOBILE INSPECTION & MAINTENANCE ROBOT

- Designed a wheeled mobile robotic platform for inspection and cleaning in confined industrial environments.
- Developed mechanical chassis and drive mechanism, integrating DC motors with motor drivers and ESP32-based control.
- Designed a servo-actuated scraping mechanism for surface cleaning and residue removal.
- Focused on applications in industrial inspection, maintenance automation, and safety monitoring.

FIXED-WING UAV PLATFORM – BOEING NATIONAL AEROMODELLING COMPETITION

- Designed and fabricated a fixed-wing UAV airframe, focusing on aerodynamic stability and structural efficiency.
- Worked on control surface mechanisms, propulsion selection, and weight optimization.
- Represented institute at Boeing National Aeromodelling Competition, achieving Top 30 national ranking.
- Gained hands-on experience in aerodynamics, flight testing, and iterative design improvements.

HOVERCRAFT – MECHANICAL DESIGN & STABILITY ANALYSIS

- Designed a hovercraft prototype, focusing on lift generation and directional stability.
- Applied principles of fluid mechanics and mechanical design to optimize airflow and skirt configuration.
- Worked on structural layout, propulsion placement, and stability considerations.
- Improved understanding of mechanical systems operating in low-friction environments.

SINGLE-AXIS HELICOPTER FOR MARS ATMOSPHERE (ISRO ROBOTICS COMPETITION)

- Designed and fabricated a single-axis rotorcraft prototype to study lift generation in low-density atmospheric conditions.
- Focused on rotor geometry, weight optimization, and mechanical stability.
- Conducted ground and functional testing under Earth conditions to validate mechanical integrity.
- Performed analytical study of aerodynamic constraints and power requirements for extraterrestrial operation.